

Week 2: Assignment 1

1. Which factors appear to influence subscription renewal the most?

From the dataset and the models, it seems that watch hours per week, monthly spending, subscription type, and number of devices used have the most effect on subscription renewal. Users who watch more content and spend more money are more likely to renew their subscriptions. These factors help in predicting whether a user will continue the subscription or not.

2. Why is subscription renewal a classification problem?

Subscription renewal is a classification problem because the output belongs to one of two categories: Yes or No. The model classifies users into these categories based on their data. Since the target variable is categorical and not numerical, classification algorithms are used.

3. Why is monthly spending a regression problem?

Monthly spending is a regression problem because the output is a numerical value. The model predicts the amount of money a user spends every month, which can take many different values. Regression algorithms are used when the target variable is continuous.

4. Which algorithm performed better for renewal prediction?

In this assignment, both Decision Tree and K-Nearest Neighbors (KNN) were used for predicting subscription renewal. The KNN model gave slightly better accuracy than the Decision Tree model. Therefore, KNN performed better for this dataset and provided more accurate predictions.

5. How could the platform use these predictions to improve customer retention?

Netflix can use these predictions to identify users who are less likely to renew their subscriptions. The company can then provide personalized recommendations, discounts, offers, or notifications to those users. This can help increase customer satisfaction and improve customer retention.